Problem Set6 Key

```
library(tidyverse)
```

```
Warning: package 'purrr' was built under R version 4.5.3
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   4.0.0     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(summarytools)
```

```
Attaching package: 'summarytools'

The following object is masked from 'package:tibble':

    view
```

1 Problem 1

The weights of 15 randomly selected girls at birth from a single hospital in New York recorded an average of 3.09 kg and standard deviation of 0.29 kg. Suppose the hospital claimed that the average weight of girls at birth is higher than 3 kg.

a. List the parameter of interest and null and alternative hypotheses for this problem. [1 pt.]

The parameter of interest is the population/long-run average weight of girls at birth. The hypotheses are:

$$H_0 : \mu = 3; H_a : \mu > 3$$

b. Calculate the test statistic. [1pt.]

```
xbar <- 3.09
mu_null <- 3
sdsamp <- 0.29
sampsize <- 15

t<- (xbar-mu_null)/(sdsamp/sqrt(sampsize))
t
```

```
[1] 1.20196
```

c. Calculate the p-value. [1 pt.]

```
df <- sampsize - 1
pt(t,df=df,lower.tail=F)
```

```
[1] 0.1246582
```

d. Do we reject or fail to reject the null hypothesis? [1 pt.]

Since significance level is not provided, use 0.05 as the significance level. We fail to reject the null hypothesis.

e. Do we have sufficient evidence to support the hospital's claim? [1pt.]

No, we have insufficient evidence to support the local hospital's claim.

2 Problem 2

The file HINTS subset.csv contains a subset of the Health information National Trends Survey 6 (HINTS 6). The households were asked if they used health or wellness apps on their tablet or smartphone (column UsedHealthWellnessApps2) and the type of community they lived in (column PR_RUCA_2010). Suppose you want to test the claim that more than 60% of the US population have used health or wellness apps on their mobile devices.

```
hints<- read.csv("datasets/HINTS subset.csv")
glimpse(hints)
```

```
Rows: 500
Columns: 4
$ X                       <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14,~
$ HHID                    <int> 23004900, 23004313, 21011500, 23022289, 210041~
$ UsedHealthWellnessApps2 <chr> "Yes", "Yes", "Yes", "Yes", "No", "No", "No", ~
$ PR_RUCA_2010            <chr> "Metropolitan", "Metropolitan", "Metropolitan"~
```

a. List the parameter of interest and null and alternative hypotheses for this problem. [1 pt.]

The parameter of interest is the proportion of the US population/population proportion who have used health or wellness apps on their mobile devices. The hypotheses are:

$$H_0 : \pi = 0.6; H_a : \pi > 0.6$$

b. Calculate the test statistic. [1pt.]

```
freq(hints$UsedHealthWellnessApps2)
```

```
Frequencies
hints$UsedHealthWellnessApps2
Type: Character

             Freq   % Valid   % Valid Cum.   % Total   % Total Cum.
---------- ------ --------- -------------- --------- --------------
         No    186     37.20          37.20     37.20          37.20
        Yes    314     62.80         100.00     62.80         100.00
      <NA>      0                                0.00         100.00
      Total    500    100.00         100.00    100.00         100.00
```

```r
sampsize <-500
phat <- 314/sampsize
pi_null <- 0.6
se <- sqrt(pi_null*(1-pi_null)/sampsize)

z <- (phat-pi_null)/se
z
```

```
[1] 1.278019
```

c. Calculate the p-value. [1 pt.]

```r
pnorm(z,lower.tail=F)
```

```
[1] 0.1006213
```

d. Do we reject or fail to reject the null hypothesis? [1 pt.]

At a 0.05 significance level, we fail to reject the null hypothesis.

e. Do we have sufficient evidence to support the claim regarding health and wellness app use? [1pt]

No, we have insufficient evidence to support the claim.

3 Problem 3

The data set in hflashes.csv contains data from a 14-year cohort study by Freeman et al (2001) that investigated the occurrence of hot flashes in 375 participants. Suppose you want to test if there is a difference between the variance of the baseline estradiol measurements (column estra) for current smokers and non-current smokers (column f1a).

```r
hflashes <- read.csv("datasets/hflashes.csv")
glimpse(hflashes)
```

```
Rows: 375
Columns: 14
$ pt       <int> 3, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 19, 20, 23, 24~
$ ageg     <int> 2, 3, 1, 1, 2, 3, 2, 2, 2, 3, 2, 1, 1, 1, 1, 3, 2, 1, 1, 1, 2~
$ aagrp    <int> 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 1, 0, 1, 1, 0, 0, 0, 0, 1, 0~
```

```
$ edu      <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 0, 1, 1, 1, 1~
$ d1       <int> 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 0, 1, 0, 1, 0~
$ f1a      <int> 0, 0, 1, 0, 1, 0, 0, 1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1~
$ pcs12    <dbl> 56.80537, 59.18338, 57.73952, 55.83575, 55.89324, NA, 55.5009~
$ hotflash <int> 0, 0, 0, 1, 0, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0~
$ bmi30    <int> 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0~
$ estra    <dbl> 106.710, 31.250, 13.410, 10.640, 24.060, 37.305, 26.320, 24.1~
$ fsh      <dbl> 3.005, 11.195, 14.545, 5.530, 9.780, 10.290, 7.960, 4.775, 7.~
$ lh       <dbl> 2.980, 5.760, 5.595, 2.260, 2.600, 3.395, 3.570, 2.095, 3.660~
$ testo    <dbl> 7.680, 11.930, 24.375, 8.280, 4.050, 8.275, 15.995, 12.340, 1~
$ dheas    <dbl> 61.225, 104.920, 117.450, 36.850, 11.165, 100.360, 76.780, 83~
```

a. List the parameter of interest and null and alternative hypotheses for this problem. [1 pt.]

The parameter of interest is the ratio of population variances of estradiol levels between smokers and non-smokers. The hypotheses are:

$$H_0 : \sigma_{ns}^2/\sigma_s^2 = 1; H_a : \sigma_{ns}^2/\sigma_s^2 \neq 1$$

b. Calculate the test statistic. [1pt.]

```
vartest <- var.test(estra~f1a,data=hflashes)
vartest
```

```
    F test to compare two variances

data:  estra by f1a
F = 1.2964, num df = 236, denom df = 137, p-value = 0.09462
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.9558288 1.7372739
sample estimates:
ratio of variances
          1.296368
```

The test statistic value is F=1.2963676.

c. Calculate the p-value. [1 pt.]

The p-value is p=0.0946195.

d. Do we reject or fail to reject the null hypothesis? [1 pt.] We fail to reject the null hypothesis.

e. Comment on the strength/sufficiency of evidence against the null hypothesis. [1pt] There is insufficient evidence to claim that there is a difference in the variance of the estradiol levels between smokers and non-smokers.

4 Problem 4

The data set in hflashes.csv contains data from a 14-year cohort study by Freeman et al (2001) that investigated the occurrence of hot flashes in 375 participants. Suppose you want to test if there is a difference between the long-run average of the baseline estradiol measurements (column estra) for current smokers and non-current smokers.

a. Based on your answer in (3d) and (3e), is it recommended to assume equal variances between the two groups? [1 pt.]

It is recommended to assume equal variances.

b. List the parameter of interest and null and alternative hypotheses for this problem. [1 pt.]

The parameter of interest is the population/long-run difference in average estradiol levels between smokers and non-smokers, The hypotheses are:

$$H_0 : \mu_{ns} - \mu_s = 0; H_a : \mu_{ns} - \mu_s \neq 0$$

c. Calculate the p-value. [1 pt.]

```
ttest <- t.test(estra~f1a,data=hflashes,var.equal=T)
ttest
```

```
    Two Sample t-test

data:  estra by f1a
t = -0.22103, df = 373, p-value = 0.8252
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
95 percent confidence interval:
 -6.644836  5.301924
sample estimates:
mean in group 0 mean in group 1
       42.67546        43.34692
```

d. Do we reject or pt. to reject the null hypothesis? [1 pt.]

We fail to reject the null hypothesis.

e. Comment on the strength/sufficiency of evidence against the null hypothesis. [1pt]

There is insufficient evidence to support the hypothesis that there is a difference in estradiol levels between smokers and non-smokers in this population.

5 Problem 5

A study was performed to investigate the association between acetaminophen use during pregnancy and the children's risk of neurodevelopmental disorders. In a retrospective study that included 2,480,797 children, 185,909 children were exposed to acetaminophen and 2,294,888 were not. Out of those who were diagnosed with autism, 5,912 children exposed to acetaminophen, while 62,672 children were not exposed to acetaminophen. Suppose we were interested in whether there is a difference in proportion of children diagnosed with autism between the different exposures to acetaminophen.

a. List the parameter of interest and null and alternative hypotheses for this problem. [1 pt.]

The parameter of interest is the long-run difference in proportion/ difference in population proportion of children diagnosed with autism between the different exposures to acetaminophen. The hypotheses are:

$$H_0 : \pi_{exposed} - \pi_{notexposed} = 0; H_a : \pi_{exposed} - \pi_{notexposed} \neq 0$$

b. Calculate the p-value. [1 pt.]

```
x1 <- 5912
x2 <- 62672
sampsize1 <- 185909
sampsize2 <- 2294888
p_pool <- (x1+x2)/(sampsize1+sampsize2)

se <- sqrt(p_pool*(1-p_pool)*(1/sampsize1+1/sampsize2))

pi1 <- x1/sampsize1
pi2 <- x2/sampsize2
```

```r
z <- (pi1-pi2)/se
z
```

```
[1] 11.35955
```

```r
# two-sided p-value
p <- pnorm(abs(z),lower.tail=F) + pnorm(-abs(z))
p
```

```
[1] 6.648664e-30
```

The p-value is $6.6486643 \times 10^{-30}$.

c. Do we reject or fail to reject the null hypothesis? [1 pt.]

We reject the null hypothesis.

d. Comment on the strength/sufficiency of evidence against the null hypothesis. [1pt]

We have sufficient evidence to support that the proportion of children diagnosed with autism after exposure to acetaminophen is not equal to the proportion of children diagnosed with autism without exposure to acetaminophen.

e. Is our result indicative of a causal relationship between acetaminophen exposure and autism prevalence in children? Why or why not? (Hint: Did we consider ALL variables that could affect both variables considered in this test?) [1pt]

NO, we cannot conclude a causal relationship between acetaminophen exposure and autism prevalence. Other possible confounding variables such as family history, social factors, and other environmental factors were not considered in the analysis that could have affected both exposure and autism diagnosis.

6 Problem 6

The data set "acupuncture.csv" includes data from the control group of a randomized controlled trial investigating the effect of acupuncture therapy on headache severity. Each participant was identified using an ID number in the id column. The headache severity was measured for each participant before receiving the treatment and at a 3-month follow-up. The column `pk1` includes the baseline headache severity score of the participants while the column `pk2` includes the headache severity score after 3 months.

```r
acu <- read.csv("datasets/acupuncture.csv")
glimpse(acu)
```

```
Rows: 173
Columns: 5
$ X     <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 1~
$ id    <int> 112, 113, 114, 130, 131, 137, 138, 141, 149, 150, 161, 169, 184,~
$ group <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
$ pk1   <dbl> 9.25, 42.50, 24.25, 21.75, 14.50, 11.75, 56.50, 15.50, 49.25, 9.~
$ pk2   <dbl> 4.75, 34.50, 16.25, 2.00, 20.00, 11.25, 70.50, 5.75, 9.75, 8.75,~
```

a. What test is the most appropriate to test whether there is a difference between the average headache severity scores of the participants before receiving the treatment and the 3-month follow-up? [1pt.]

Paired t-test because these scores were measured from the same participant.

b. What are the statistical hypotheses for testing whether the average headache severity score in this cohort was lower at the 3-month follow-up compared to the baseline? [2pts.]

$$H_0 : \mu_d = 0; H_a : \mu_d > 0$$

where μ_d is defined as the average difference between baseline and 3-month follow-up.

> **!** Important
>
> If μ_d is the difference between the treatment between 3-month follow-up and baseline, then the hypothesis should be:
>
> $$H_0 : \mu_d = 0; H_a : \mu_d < 0$$

c. What is the value of the test statistic and the corresponding number of degrees of freedom? [2pts.]

```r
ttest <- t.test(acu$pk1,acu$pk2,paired=T,alternative="greater")
ttest
```

```

    Paired t-test

data:  acu$pk1 and acu$pk2
```

```
t = 6.9415, df = 172, p-value = 3.799e-11
alternative hypothesis: true mean difference is greater than 0
95 percent confidence interval:
 4.916204      Inf
sample estimates:
mean difference
       6.453757
```

```r
# OR

ttest <- t.test(acu$pk2,acu$pk1,paired=T,alternative="less")
ttest
```

```
    Paired t-test

data:  acu$pk2 and acu$pk1
t = -6.9415, df = 172, p-value = 3.799e-11
alternative hypothesis: true mean difference is less than 0
95 percent confidence interval:
      -Inf -4.916204
sample estimates:
mean difference
      -6.453757
```

d. What is the p-value? [1pt.]

The p-value is $3.7991629 \times 10^{-11}$.

e. At a significance level of 0.001, do we have sufficient evidence that the average headache severity score in this cohort was lower at the 3-month follow-up compared to the baseline? [1pt.]

Yes, we have sufficient evidence that the average headache severity score in this cohort was lower at the 3-month follow-up compared to the baseline.